

Tutorial – 04

1. The employee list for a company contains employee code, name, designation and basic pay. The employee is given a house rent allowance (HRA) of 10% of the basic pay and dearness allowance (DA) of 45% of the basic pay. The total pay of the employee is calculated as Basic Pay + HRA + DA. Write a class to define the details of the employee. Write a constructor to assign the required initial values. Add a method to calculate HRA, DA and total pay and print them. Write another class with the main method. Create objects for three different employees and calculate HRA, DA and total pay.

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    class Employee4
```

```
    {
```

```
        private int employeeCode;
```

```
        private string name;
```

```
        private string designation;
```

```
        private double basicPay;
```

```
        private double hra;
```

```
        private double da;
```

```
        private double totalPay;
```

```
        //constructor to initialize employee details
```

```
        public Employee4(int code, string empName, string desig, double pay)
```

```
        {
```

```
            employeeCode = code;
```

```
            name = empName;
```

```
            designation = desig;
```

```
            basicPay = pay;
```

```
        }
```

```
        //method to calculate and display hra, da and total pay
```

```
        public void calculate()
```

```
        {
```

```
            hra = basicPay * 0.20;
```

```
            da = basicPay * 0.10;
```

```
            totalPay = basicPay + hra + da;
```

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```
}  
public void display()  
{  
    Console.WriteLine("\nEmployee Details:");  
    Console.WriteLine("Code: " + employeeCode);  
    Console.WriteLine("Name: " + name);  
    Console.WriteLine("Designation: " + designation);  
    Console.WriteLine("Basic Pay: " + basicPay);  
    Console.WriteLine("HRA(10%): " + hra);  
    Console.WriteLine("DA(45%): " + da);  
    Console.WriteLine("Total Pay: : " + totalPay);  
}  
}  
class T4_01  
{  
    public static void T4_01Main()  
    {  
        Employee4 s = new Employee4(101, "Tisha", "Developer", 50000);  
        s.calculate();  
        s.display();  
        Employee4 s1 = new Employee4(102, "Tisha", "Developer", 50000);  
        s1.calculate();  
        s1.display();  
    }  
}  
}
```

Output:

```
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----Employee Details----  
Code: 101  
Name: Tisha  
Designation: Developer  
Basic Pay: 50000  
HRA(10%): 10000  
DA(45%): 5000  
Total Pay: 65000  
  
----Employee Details----  
Code: 101  
Name: Tisha  
Designation: Developer  
Basic Pay: 50000  
HRA(10%): 10000  
DA(45%): 5000  
Total Pay: 65000
```

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2. From the following code and given output, complete missing statements and find out error code and correct it.

Solution:

//Base class or Parent class.

class Shape

{

 public double Width;

 public double Height;

 public void ShowDim()

 {

 Console.WriteLine("Width and height are " + Width + " and " + Height);

 }

}

// Triangle is derived from Shape.

//Derived class or Child class.

class Triangle : Shape

{

 public string Style; // style of triangle

 // Return area of triangle.

 //.....Missing statement-1.....//

 {

 return Width * Height / 2;

 }

 // Display a triangle's style.

 public void ShowStyle()

 {

 //.....Missing statement-2.....//

 }

}

//Driver class which runs the program.

class Driver

{

 static void Main()

 {

 Triangle t1 new Triangle();

 Triangle t2 new Triangle();

 t1.Width =4.0;

 t1.Height =4.0;

 t1.Style ="isosceles";

 t2.Width =8.0;

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```
t2.Height =12.0;
t2.Style ="right";
Console.WriteLine("Info for t1: ");
t1.ShowStyle();
t1.ShowDim();
Console.WriteLine("Area is " + t1.Area());
Console.WriteLine();
Console.WriteLine("Info for t2: ");
t2.ShowStyle();
t2.ShowDim();
Console.WriteLine("Area is " + t2.Area());
}
```

Output

Info for t1:

Triangle is isosceles

Width and height are 4 and 4

Area is 8

Info for t2:

Triangle is right

Width and height are 8 and 12

Area is 48

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
// Base class or Parent class
```

```
class Shape4
```

```
{
```

```
    public double Width;
```

```
    public double Height;
```

```
    public void ShowDim()
```

```
    {
```

```
        Console.WriteLine("Width and height are " + Width + " and " + Height);
```

```
    }
```

```
}
```

```
// Triangle is derived from Shape (Child class)
```

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```
class Triangle4 : Shape4
{
    public string Style; // Style of triangle

    // Missing statement-1: Method signature for calculating area
    public double Area()
    {
        return Width * Height / 2;
    }

    // Display a triangle's style
    public void ShowStyle()
    {
        // Missing statement-2: Printing the style property
        Console.WriteLine("Triangle is " + Style);
    }
}
```

```
class T4_02
{
    public static void T4_02Main()
    {
        Triangle4 t1 = new Triangle4();
        Triangle4 t2 = new Triangle4();

        t1.Width = 4.0;
        t1.Height = 4.0;
        t1.Style = "isosceles";

        t2.Width = 8.0;
        t2.Height = 12.0;
        t2.Style = "right";

        Console.WriteLine("Info for t1: ");
        t1.ShowStyle();
        t1.ShowDim();
        Console.WriteLine("Area is " + t1.Area());

        Console.WriteLine("\nInfo for t2: ");
        t2.ShowStyle();
        t2.ShowDim();
    }
}
```

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```
Console.WriteLine("Area is " + t2.Area());
```

```
    }  
}  
}
```

Output:

```
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Info for t1:  
Triangle is isosceles  
Width and height are 4 and 4  
Area is 8  
  
Info for t2:  
Triangle is right  
Width and height are 8 and 12  
Area is 48
```

3. Draw a real picture for single level inheritance. Perform following tasks.

Task 1: Create a class

Task 2: Add few data members as private, protected and public

Task 3: Add few methods as public to work on defined data members

Task 4: Create another applicable class which inherits members from above class

Task 5: Add few data members as private, protected and public into second class

Task 6: Add few methods as public to work on defined data members into second class

Task 7: Create a Demo class with the main method.

Task 8: Create at least two objects of a second class defined in Task 4 into the main method and call all methods using that object.

Task 9: Write a comment for each important portion of code like data members' declaration, methods, some important logic etc.

Task 10: Summarize the above solution in your own few words to visualize the solution to the end user.

Code:

```
using System;
```

```
namespace Practicals
```

```
{
```

```
    //base class
```

```
    class Person4
```

```
    {
```

```
        private string name;
```

```
        protected int age;
```

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```
public string address;
//constructor
public Person4(string nm, int ag, string addr)
{
    name = nm;
    age = ag;
    address = addr;
}
public void ShowName()
{
    Console.WriteLine("Name: " + name);
}
public void ShowAge()
{
    Console.WriteLine("Age: " + age);
}
public void ShowAddress()
{
    Console.WriteLine("Address: " + address);
}
}
//derived class
class Student4 : Person4
{
    private int rollNo;
    protected string branch;
    public double marks;
    public Student4(string name, int age, string address, int roll, string br, double m) : base(name,
age, address)
    {
        rollNo = roll;
        branch = br;
        marks = m;
    }
    public void ShowRollNo()
    {
        Console.WriteLine("Roll No: " + rollNo);
    }
    public void ShowBranch()
    {
        Console.WriteLine("Branch: " + branch);
    }
}
```

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```
}  
public void ShowMarks()  
{  
    Console.WriteLine("Marks: " + marks);  
}  
  
//method to call all student details  
public void DisplayStudent()  
{  
    ShowName();  
    ShowAge();  
    ShowAddress();  
    ShowRollNo();  
    ShowBranch();  
    ShowMarks();  
}  
}  
class T4_03  
{  
    public static void T4_03Main()  
    {  
        Student4 s1= new Student4("Tisha", 19, "Porbandar", 46, "CE", 89);  
        s1.DisplayStudent();  
        Student4 s2= new Student4("Titu", 20, "Bhuj", 38, "CE", 90);  
        s2.DisplayStudent();  
    }  
}
```

Output:

```
Mandaviya Tisha Jayesh - 25SOECE13066  
Name: Tisha  
Age: 19  
Address: Porbandar  
Roll No: 46  
Branch: CE  
Marks: 89  
Name: Titu  
Age: 20  
Address: Bhuj  
Roll No: 38  
Branch: CE  
Marks: 90
```

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4. From the following code and given output complete missing statements and find out error code and correct it.

```
using System;
namespace StaticVarApplication
{
    class StaticVar
    {
        public static int num;
        public counting()
        {
            num++;
        }
        //.....Missing statement.....//
        {
            return num;
        }
    }
    class StaticTester
    {
        static void Main(string[] args)
        {
            StaticVar s = new StaticVar();
            s.count();
            s.count();
            s.count();
            Console.WriteLine("Variable num: {0}", StaticVar.getNum());
            Console.ReadKey();
        }
    }
}
```

Output:

Variable num: 3

Code:

using System;

namespace Practicals

```
{
    class StaticVar
    {
        public static int num;
```

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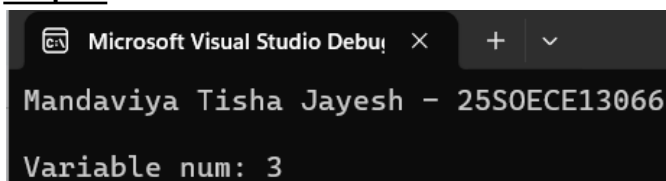
// Corrected from 'counting()' to match the method call 's.count()'

```
public void count()
{
    num++;
}
public static int getNum()
{
    return num;
}
```

```
class T4_04
{
    public static void T4_04Main()
    {
        StaticVar s = new StaticVar();
        s.count();
        s.count();
        s.count();

        Console.WriteLine("Variable num: {0}", StaticVar.getNum());
    }
}
```

Output:



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Variable num: 3

5. Find out error code and correct it. Print appropriate output as desired.

```
using System;
public class A // This is the base class.
{
    public A(int value)
    {
        // Executes some code in the constructor.
        Console.WriteLine("Base constructor A()");
    }
}
```

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```
public class B : A // This class derives from the previous class.
{
    public B(int value) : base(value)
    {
        // The base constructor is called first.
        // ... Then this code is executed.
        Console.WriteLine("Derived constructor B()");
    }
}
class Program
{
    static void Main()
    {
        // Create a new instance of class A, which is the base class.
        // ... Then create an instance of B, which executes the base constructor.
        A a = A(0);
        B b = B(1);
    }
}
```

Expected Outcome :

Base constructor A()

Base constructor A()

Derived constructor B()

Code:

using System;

namespace Practicals {

```
public class A
{
    public A(int value)
    {
        // Executes some code in the constructor.
        Console.WriteLine("Base constructor A(value: {0})", value);
    }
}
public class B : A // This class derives from the previous class.
{
    public B(int value) : base(value)
    {

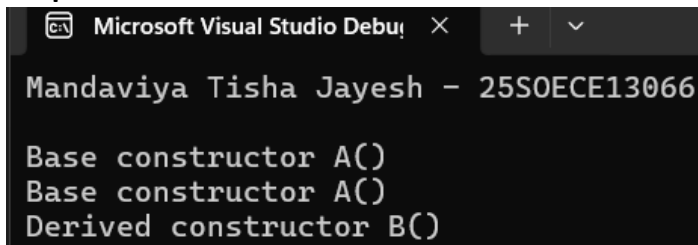
```

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```
// The base constructor is called first.  
// ... Then this code is executed.  
Console.WriteLine("Derived constructor B(value: {0})", value);
```

```
}  
}  
class T4_05  
{  
    public static void T4_05Main()  
    {  
        // Create a new instance of class A, which is the base class.  
        // ... Then create an instance of B, which executes the base constructor.  
        A a = new A(0);  
        B b = new B(1);  
    }  
}
```

Output:



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Base constructor A()
Base constructor A()
Derived constructor B()

6. Find out the error code and correct it. Print appropriate output as desired. Solution:

```
using System;  
abstract class Test  
{  
    int a;  
    abstract void A();  
}  
class Example1 : Test  
{  
    public override void A()  
    {  
        Console.WriteLine("Example1.A");  
        base.a++;  
    }  
}  
class Example2 : Test
```

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```
{
    public override void A()
    {
        Console.WriteLine("Example2.A");
        base.a--;
    }
}
class Program
{
    static void Main()
    {
        // Reference Example1 through Test type.
        Test test1 = new Example1();
        test1.A();
        // Reference Example2 through Test type.
        Test test2 = new Example2();
        test2.A();
    }
}
```

Expected Outcome :

Example1.A

Example2.A

Code:

using System;

namespace Practicals

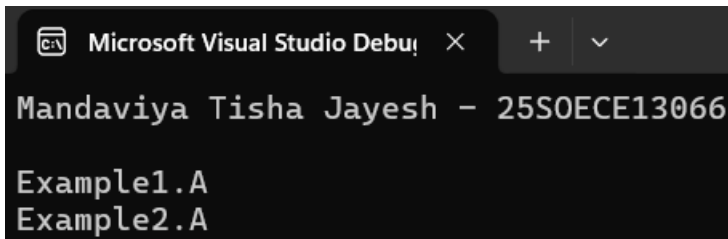
```
{
    abstract class Test
    {
        protected int a;
        public abstract void A();
    }

    class Example1 : Test
    {
        public override void A()
        {
            Console.WriteLine("Example1.A");
            a++;
        }
    }
}
```

```
class Example2 : Test
{
    public override void A()
    {
        Console.WriteLine("Example2.A");
        a--;
    }
}

class T4_06
{
    public static void T4_06Main()
    {
        // Reference Example1 through Test type.
        Test test1 = new Example1();
        test1.A();
        // Reference Example2 through Test type.
        Test test2 = new Example2();
        test2.A();
    }
}
```

Output:



Microsoft Visual Studio Debug Console window showing the output of the program. The window title is "Mandaviya Tisha Jayesh - 25SOECE13066". The output text is:

```
Example1.A
Example2.A
```

7. Refer to the given output and find out the error code and correct it.

```
sealed class A
{
    public int x;
    public int y;
}
class B : A
{
    public int z;
```

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```
}  
class SealedTest2  
{  
    static void Main()  
    {  
        A sc = new A();  
        sc.x = 110;  
        sc.y = 150;  
        Console.WriteLine("x = {0}, y = {1}", sc.x, sc.y);  
    }  
}
```

Output

x = 110, y = 150

Code:

```
using System;
```

```
namespace Practicals
```

```
{  
    class Ab  
    {  
        public int x;  
        public int y;  
    }  
}
```

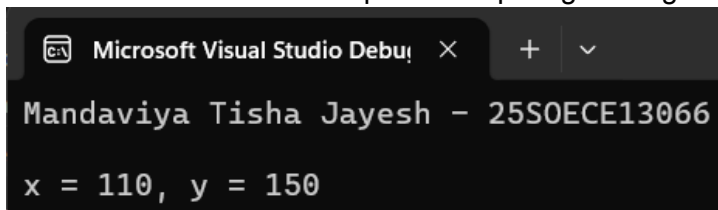
```
class Ba : Ab  
{  
    public int z;  
}
```

```
class T4_07  
{  
    public static void T4_07Main()  
    {  
        Ab sc = new Ab();  
        sc.x = 110;  
        sc.y = 150;  
        Console.WriteLine("x = {0}, y = {1}", sc.x, sc.y);  
    }  
}
```

Output:

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```
Microsoft Visual Studio Debug Console
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x = 110, y = 150
```

8. Find out the error code and correct it. Print appropriate output as desired. Solution:

```
class X
{
    protected virtual void F() { Console.WriteLine("X.F"); }
    protected virtual void F2() { Console.WriteLine("X.F2"); }
}
class Y : X
{
    sealed protected override void F() { Console.WriteLine("Y.F"); }
    protected override void F2() { Console.WriteLine("Y.F2"); }
}
class Z : Y
{
    // Overriding F
    protected override void F() { Console.WriteLine("Z.F"); }
    // Overriding F2
    protected override void F2() { Console.WriteLine("Z.F2"); }
}
class SealedMethodTest
{
    static void Main()
    {
        X Obj1 = new X();
        Obj1.F();
        Obj2.F2();
        Y Obj2 = new Y();
        Obj2.F();
        Obj2.F2();
        Z Obj3 = new Z();
        Obj3.F();
        Obj3.F2();
    }
}
```

Expected Outcome :

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X.F

X.F2

Y.F

Y.F2

Y.F

Z.F2

Code:

```
using System;
```

```
namespace Practicals
```

```
{  
    class X  
    {  
        public virtual void F() { Console.WriteLine("X.F"); }  
        public virtual void F2() { Console.WriteLine("X.F2"); }  
    }  
  
    class Y : X  
    {  
        public override void F() { Console.WriteLine("Y.F"); }  
        public override void F2() { Console.WriteLine("Y.F2"); }  
    }  
  
    class Z : Y  
    {  
        // Overriding F  
        public override void F() { Console.WriteLine("Z.F"); }  
        // Overriding F2  
        public override void F2() { Console.WriteLine("Z.F2"); }  
    }  
  
    class T4_08  
    {  
        public static void T4_08Main()  
        {  
            X Obj1 = new X();  
            Obj1.F();  
            Obj1.F2();  
  
            Y Obj2 = new Y();  
            Obj2.F();  
        }  
    }  
}
```

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```
Obj2.F2();
```

```
Z Obj3 = new Z();
```

```
Obj3.F();
```

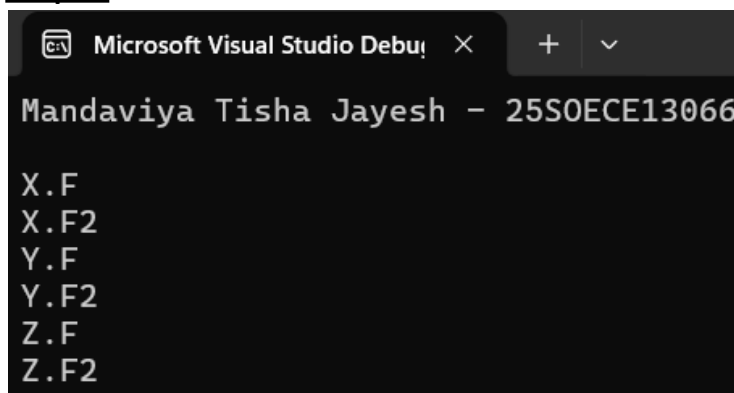
```
Obj3.F2();
```

```
}
```

```
}
```

```
}
```

Output:



```
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X.F
X.F2
Y.F
Y.F2
Z.F
Z.F2
```

9. This program will throw an exception. Add try, catch and finally blocks to handle this exception.

```
using System;
```

```
class MyClient
```

```
{
```

```
    public static void Main()
```

```
    {
```

```
        int x = 0;
```

```
        int div = 100/x;
```

```
        Console.WriteLine(div);
```

```
    }
```

```
}
```

Expected Outcome :

Cannot divide by zero.

Finally block executed.

Code:

```
using System;
```

```
namespace Practicals {
```

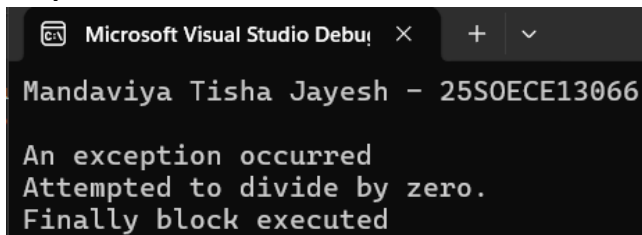
```
    class T4_09
```

```
    {
```

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```
public static void T4_09Main()
{
    try
    {
        int x = 0;
        int div = 100 / x;
        Console.WriteLine(div);
    }
    catch (Exception e)
    {
        Console.WriteLine("An error occurred: " + e.Message);
    }
    finally
    {
        Console.WriteLine("Finally block executed.");
    }
}
```

Output:



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An exception occurred
Attempted to divide by zero.
Finally block executed

10. Arrange the code to get desirable output:

```
class MyException : Exception
{
}
class MyClient
{
    public static void Main()
    {
        try
        {
            throw new MyException("my exception generated.");
        }
        catch(Exception e)
        {
        }
    }
}
```

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```
{  
}  
    Console.WriteLine("LAST STATEMENT");  
}  
  
    using System;  
    public MyException(string str)  
    {  
        Console.WriteLine("User defined exception");  
        Console.WriteLine("Exception caught here: " + e.ToString());  
    }  
}
```

Output:

Exception caught here: my exception generated.

LAST STATEMENT

Code:

using System;

namespace Practicals

```
{  
    class MyException : Exception  
    {  
        public MyException(string str)  
        {  
            Console.WriteLine("User defined exception");  
        }  
    }  
    class T4_10  
    {  
        public static void T4_10Main()  
        {  
            try  
            {  
                throw new MyException("my exception generated.");  
            }  
            catch (Exception e)  
            {  
                Console.WriteLine("Exception caught here: " + e.Message);  
            }  
            Console.WriteLine("LAST STATEMENT");  
        }  
    }  
}
```

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}

Output:

```
Microsoft Visual Studio Debu  X  +  v
Mandaviya Tisha Jayesh - 25SOECE13066
User defined exception
Exception caught here: Exception of type 'Practicals.MyException' was thrown.
LAST STATEMENT
```